



Breaking the Link Between Legal Access to Alcohol and Motor Vehicle Accidents: Evidence from New South Wales

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Research Question

How the legal access to alcohol affect:

- *Drinking Behavior*
- *Motor Vehicle Accidents*
- *Hospitalization*

Why Is It Important?

- High way funding and higher MLDA (minimum legal drinking age).
- Critiques on MLDA and other responsibilities bestowed upon 18 year olds.
- To properly craft policy, need to weigh the costs and benefits of the wide variety of alcohol-control options and to determine how the effects of these policies may complement or substitute for one another.

Why New South Wales in Australia?

Among the world leaders in their efforts against drunk driving

- *NATURAL EXPERIMENT*—The BAC (blood alcohol concentration) in limit for drivers remains at zero (since 2004, 0.02 earlier) through the MLDA threshold.
- *MORE SEVERE EFFECTS*—the MLDA is 18, and we tend to expect youths to become more responsible between the ages of 18 and 21.
- *GRADUATED LICENSING SYSTEM (GLS)*—learner's → first provisional → second provisional

Previous Findings

Mixed

- *NOT EFFECTIVE*—Dee and Evans (2001) and Carpenter (2006)
- *EFFECTIVE*—Eisenberg (2003) and Voas et al. (2003)

Identification Strategy

- Sharp RD
- Empirical Strategy

$$y_i = \beta_0 + \beta_1 MLDA + f(age_i) + \varepsilon_i \quad (1)$$

- y_i : Drinking Behavior
Motor Vehicle Accidents (MVAs)
Hospitalizations
- $MLDA$ takes a value of 1 if $age_i \geq 18$
- $f(age_i)$: interacted local linear specification, allowing the slopes to vary on each side of the threshold
- rectangular kernel weights
- A range of bandwidths including IK optimal bandwidth
- cluster standard error on age (in days)

Data

- Drinking behavior : 2001-2011
 - the Household, Income and Labor Dynamics Australia (HILDA) survey
 - indicators for drinking frequency
 - proportions of days with drinking / binge drinking



Data

- Driver and MVAs : July 2000 – December 2010
 - CrashLink and NSW Center for Road Safety
 - Crash rates for drivers are those who first obtained a provisional license between 17th and 18th birthday and after June 2000.
 - At least one vehicle was towed away or one of the occupants was injured
 - Serious accidents : e.g. drivers is injured, occur at night, the driver died
 - MVAs weighted by fatality risk: estimate fatality risk based on the predicted probability from a probit model in which fatality is a function of a set of driver and crash characteristics.

Data

- Hospital Inpatient data : July 2001 – June 2010
 - The National Hospital Morbidity Database (NHMD)
 - Inpatient episodes per 10,000 person-years
 - Alcohol intoxication/ poisoning ,Drivers injured in Motor Vehicle Accidents ,Assaults, Motorcycle accidents, Other external causes

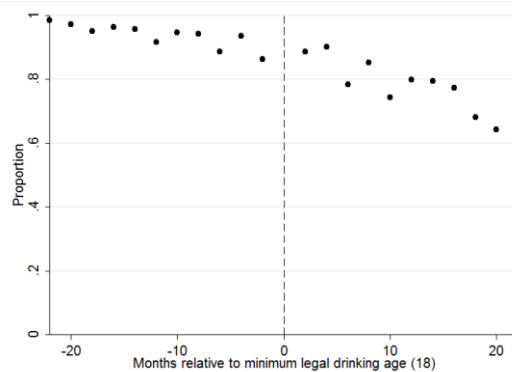
A Valid RD Design?

- Identifying assumptions: characteristics related to outcomes of interest vary smoothly through the treatment threshold.
- No manipulation of running variable: no control over age
- “Birthday Effect”
 - Outcomes may change around the time an individual’s age changes because of celebrating birthday.
 - “Donut RD” : omit observations that are in the immediate vicinity of the treatment threshold
 - One-month donut: drinking behavior
 - Seven-day donut: MVAs, Hospitalizations

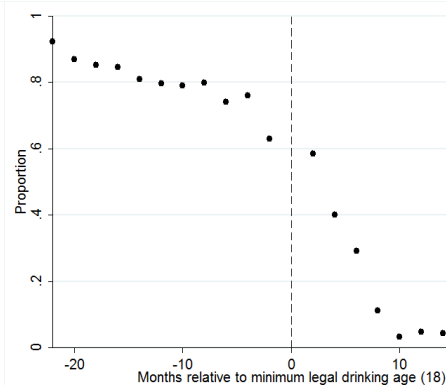
A Valid RD Design?

- Population characteristics:

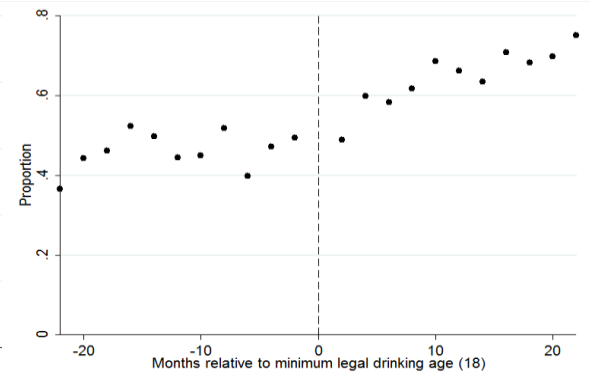
Lives with Parents



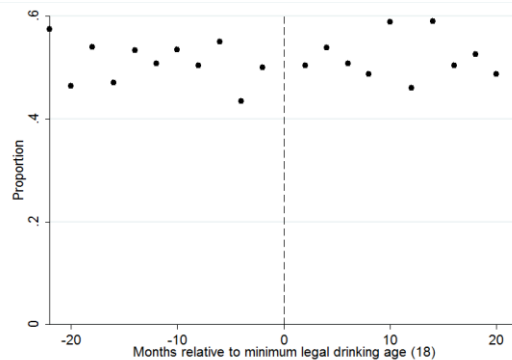
Enrolled in School



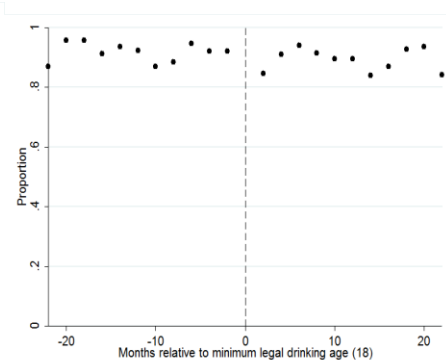
Employed



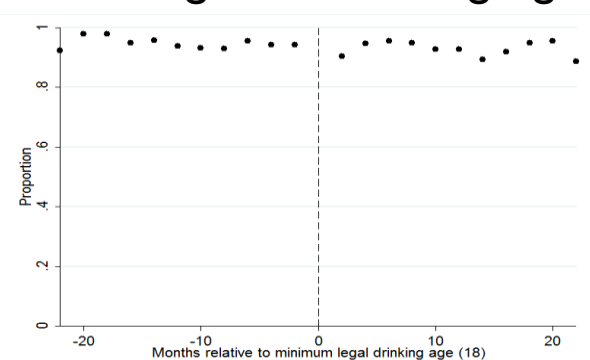
Female



Born in Australia



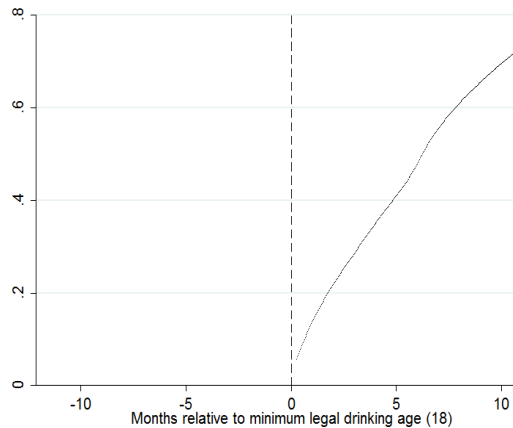
English is 1st Language



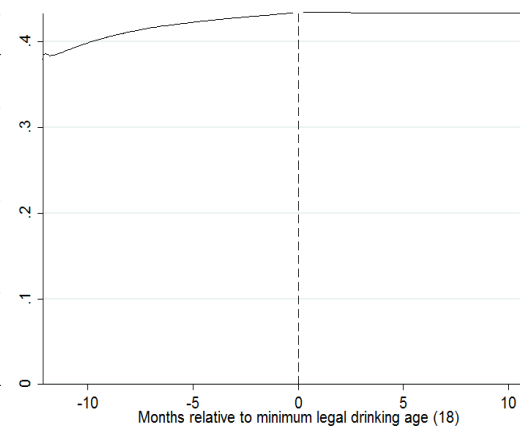
A Valid RD Design?

- Driver characteristics:

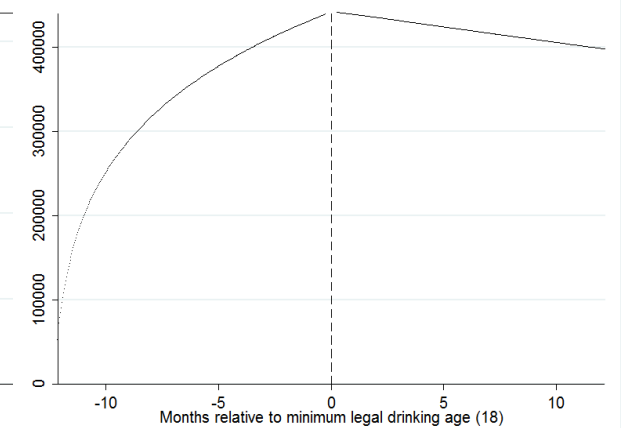
Proportion with P2 License



Proportion Female



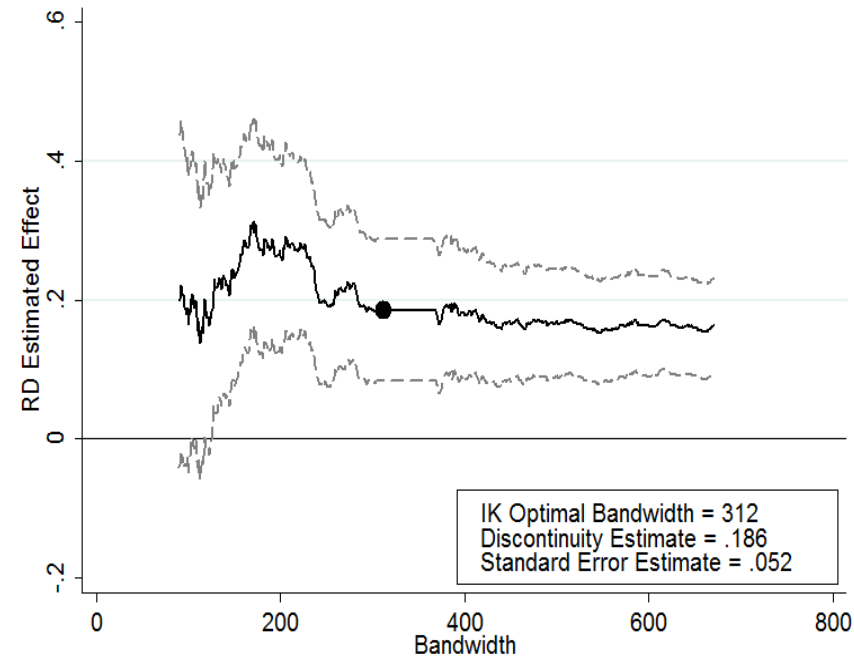
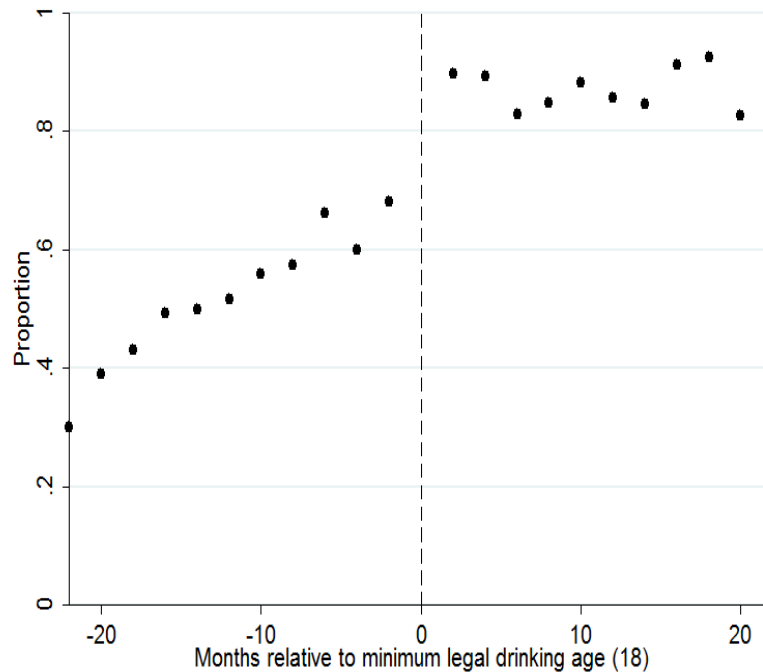
number of drivers



- Bias estimates of the effect of MLDA on MVAs upward
- Upper bounds

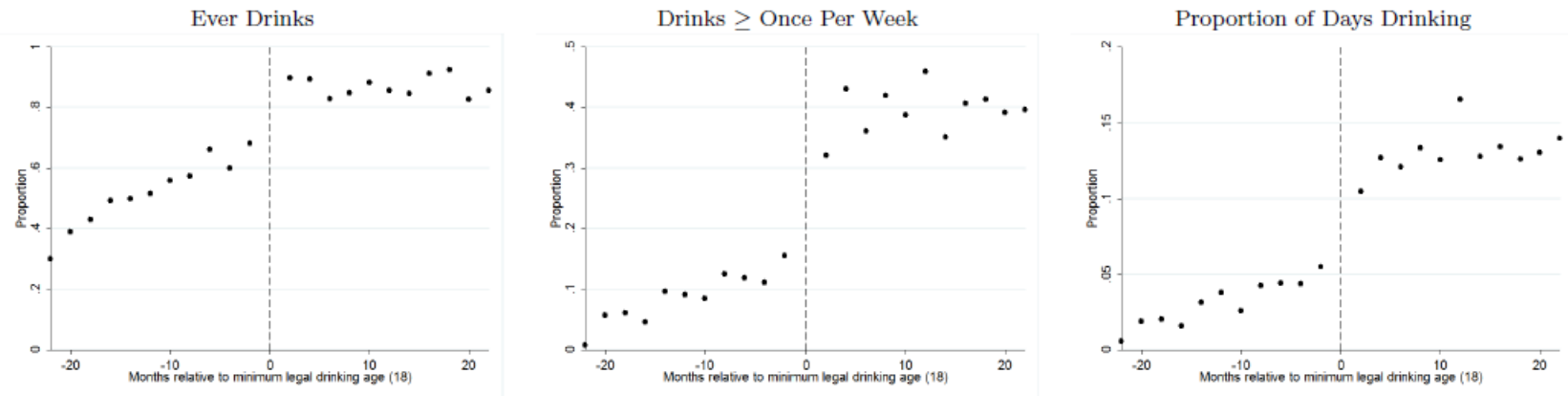
Results I—Effect on Drinking Behavior

- Ever drinks

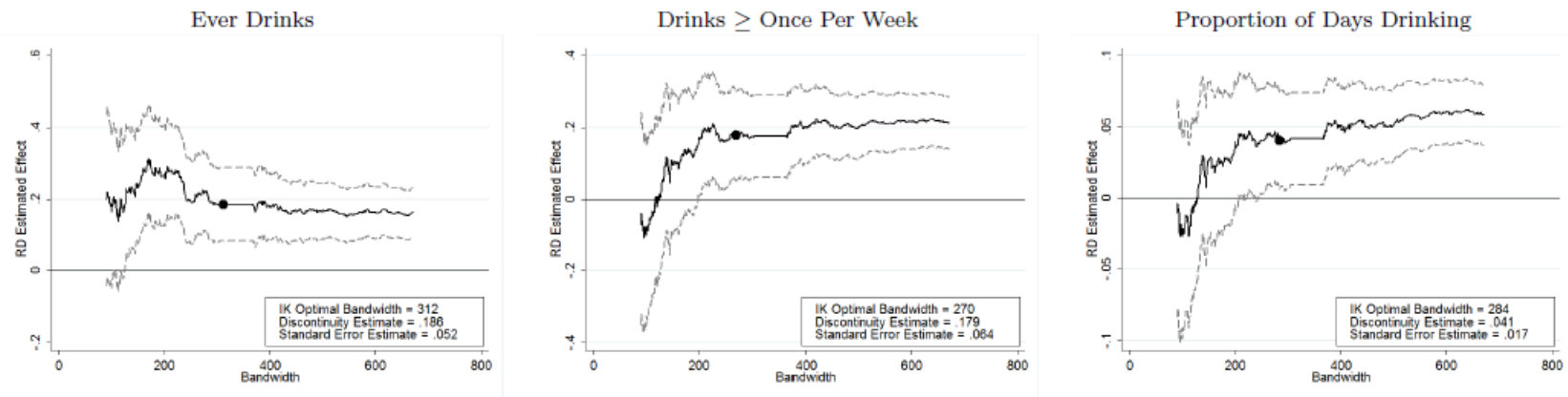


- IK optimal Bandwidth=312
- Discontinuity Estimate=0.186***

Results I—Effect on Drinking Behavior

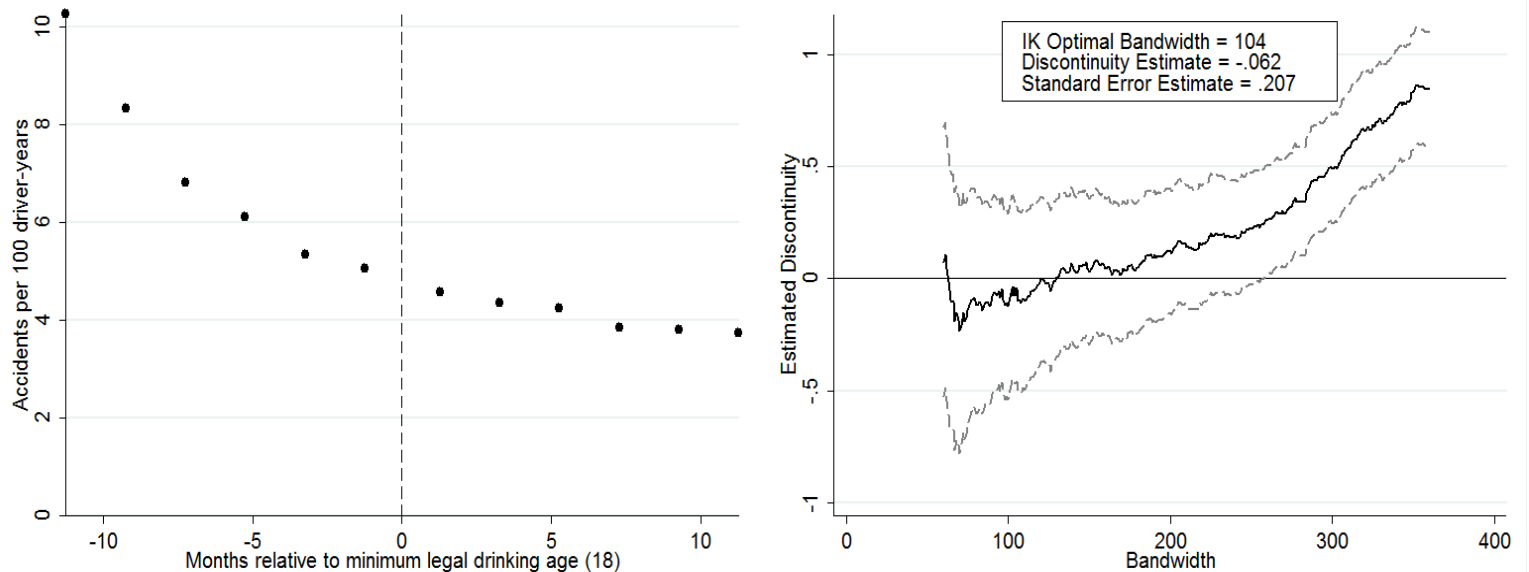


Panel B: Discontinuity Estimates By Bandwidth (In Days)



Result 2—Effect on Motor Vehicle Accidents

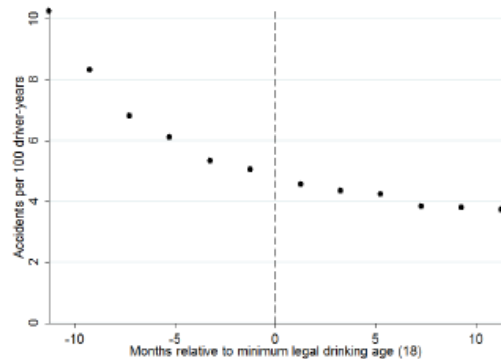
- Accident with Injury or Towed Vehicle



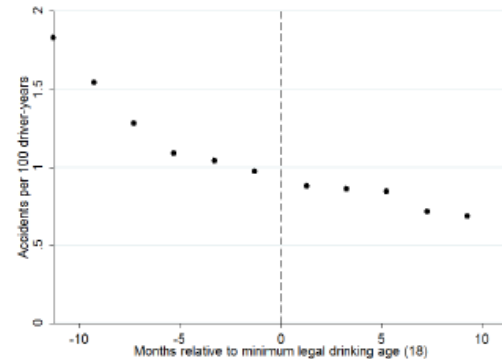
- IK Optimal Bandwidth=104
- Discontinuity Estimate=-0.062

Result 2—Effect on Motor Vehicle Accidents

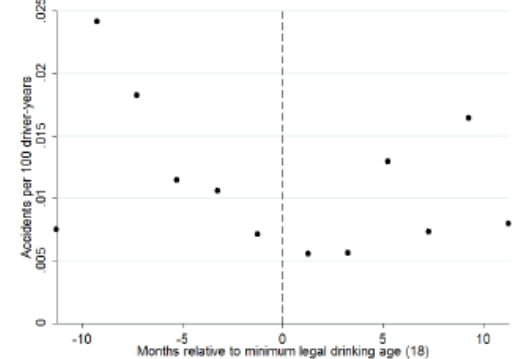
Accident with Injury or Towed Vehicle



Injured

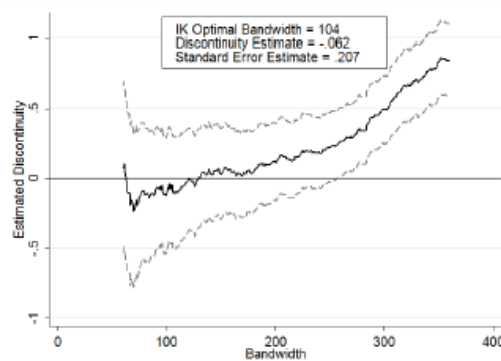


Killed

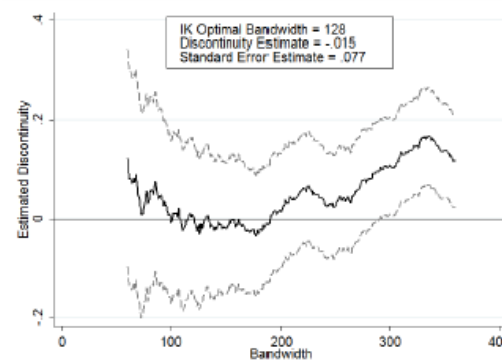


Panel B: Discontinuity Estimates By Bandwidth (In Days)

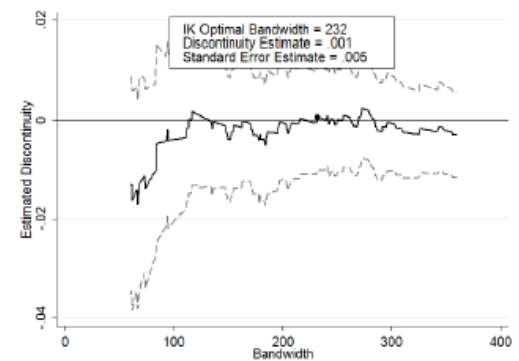
Accident with Injury or Towed Vehicle



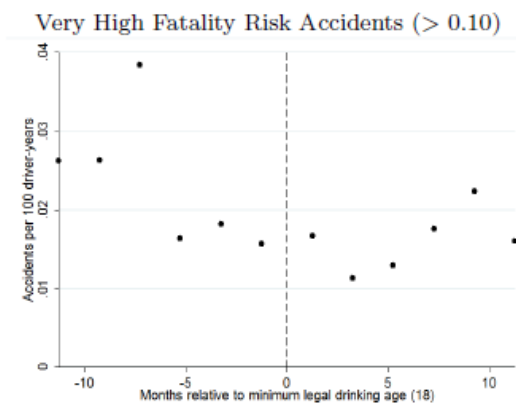
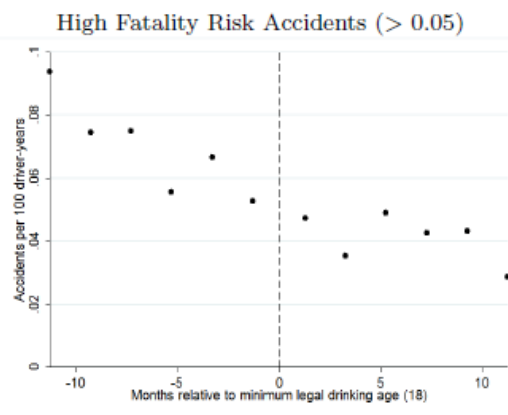
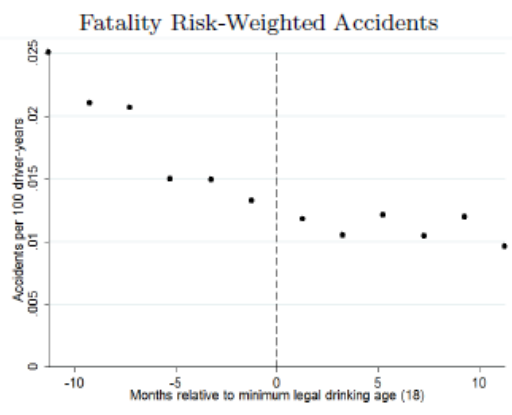
Injured



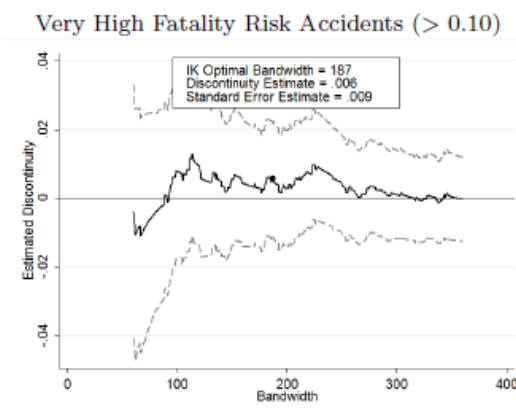
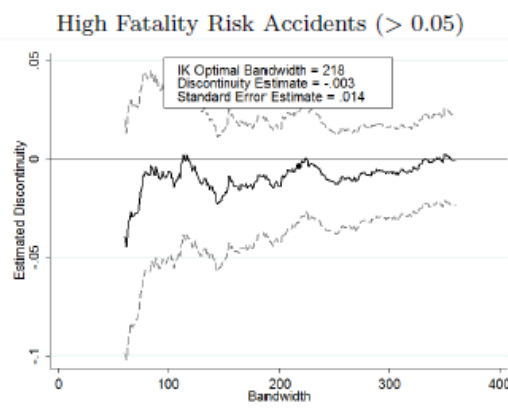
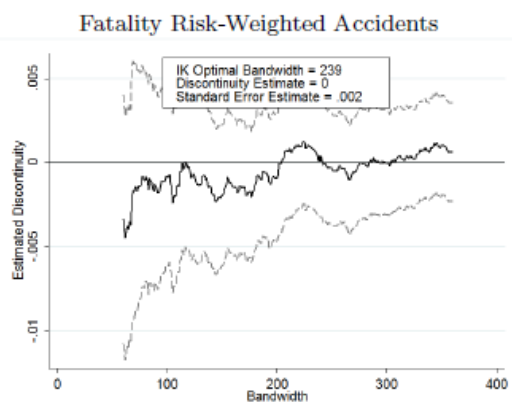
Killed



Result 2—Effect on Motor Vehicle Accidents (Weighted)

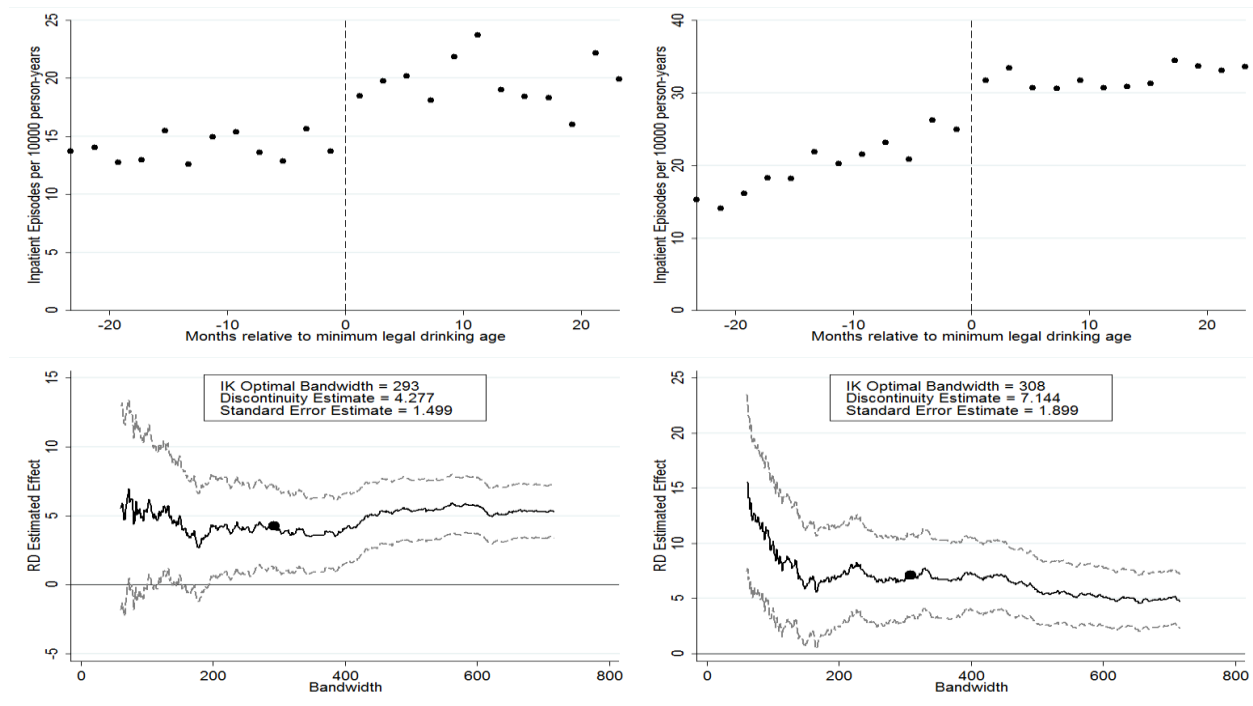


Panel B: Discontinuity Estimates By Bandwidth (In Days)



Result 3—Effect on Hospitalization

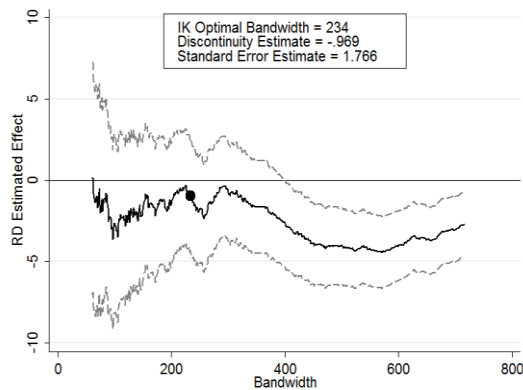
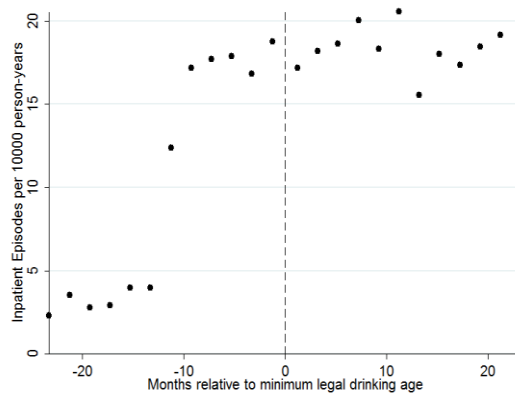
- Hospitalizations for Alcohol Poisoning and Assault



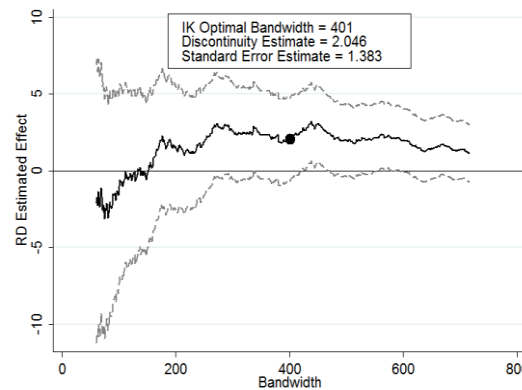
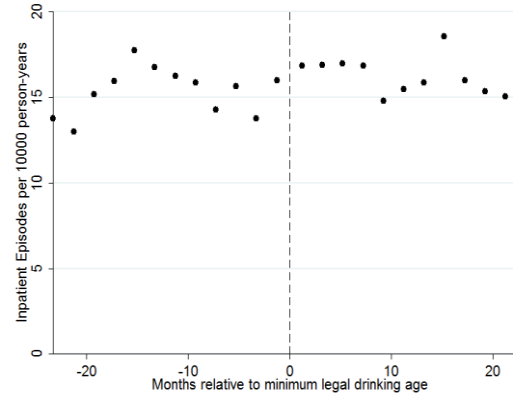
- IK Optimal Bandwidth=293; 308
- Discontinuity Estimate=4.277*** ; 7.144***

Result 3—Effect on Hospitalization

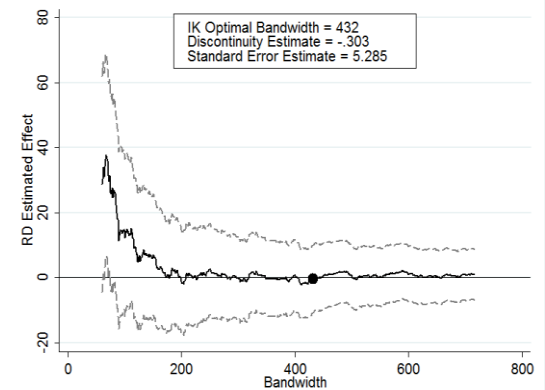
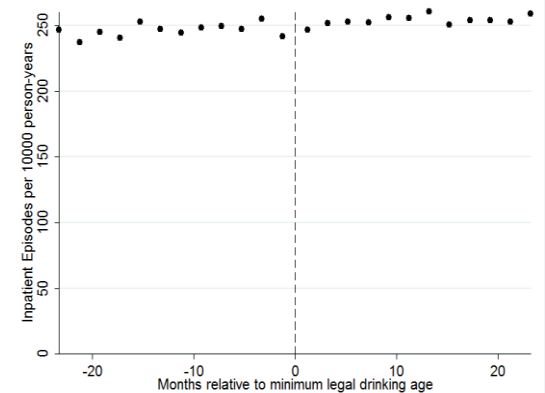
MVAs



Motorcycle Accidents



Other External Causes



Conclusion

- Youths increase drinking as they grow older.
- There is no evidence that legal access to alcohol affects motor vehicle accidents..
- There are significant effects on hospitalizations due to alcohol poisoning and assault victimization.

Concerns

- I. Does the BAC limit change in 2004 affect the outcome?
 - BAC limit for both learners and drivers with provisional licenses: it was set at 0.02 initially and reduced to zero in 2004.
 - Will both underage drinkers and legal drinkers be more concerned with their drinking behavior ?
 - Add a dummy for 2004 to capture the policy change effect

Concerns

- II. Any strong link between “drinking” and “driving after drinking”?
 - Availability of random breath test data
 - RD design to test the discontinuity in “driving after drinking” behavior.